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Acoustic Performance Comparison of Metal-Finish Wall Systems Between Adjacent Hotel Guest Rooms Using ODEON

A Geometrical-Acoustics Simulation Study of Three Separating-Wall Assemblies

Technical Report

Course: BSTC-582 Acoustical Modeling

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Executive Summary

Noise transmitted between adjacent hotel guest rooms can compromise acoustic privacy, comfort and sleep quality. This report uses the room-acoustics simulation program ODEON to evaluate how the choice of separating wall governs that transfer between two adjoining guest rooms in a hotel in Dammam, Saudi Arabia.

A single two-room model was built with a fixed sound source in the first room and two analysed receivers: R1 in the source room and R4 in the receiving room. Three separating-wall assemblies, all sharing a metal architectural finish, were compared in turn while every other parameter was held constant: Case 1, a single 1 mm aluminium sheet; Case 2, a 103 mm aluminium–rockwool–aluminium sandwich; and Case 3, a 77 mm metal-finish insulated stud partition with a rock-wool-filled cavity.

For each case the A-weighted sound-pressure level $SPL(A)$ and the octave-band spectrum were extracted at both receivers, and the A-weighted sound-level reduction between the rooms, $\Delta L_A = SPL(A)_{R1} - SPL(A)_{R4}$, was computed as the headline metric. The three cases delivered reductions of 18.4, 33.0 and 40.4 dB(A) respectively, giving predicted transmitted levels at the receiving-room receiver R4 of 40.5, 25.9 and 18.5 dB(A) from an identical 58.9 dB(A) source-room level.

Case 3 provides the best overall privacy, the strongest mid- and high-frequency control and the closest match to the intended metal-finish design, and is therefore recommended. All transmission-loss inputs are clearly identified as engineering estimates; the study is a comparative simulation and not a substitute for laboratory or field certification.

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List of Abbreviations

Abbreviation	Meaning
dB	Decibel (logarithmic unit of sound level)
dB(A)	A-weighted decibel
Hz	Hertz (cycles per second)
SPL	Sound-pressure level
SPL(A)	A-weighted sound-pressure level
TL / R	Transmission loss / sound-reduction index of a building element
Δ LA	A-weighted sound-level reduction between the source and receiving rooms
ODEON	Geometrical room-acoustics simulation software
P1	Sound-source position in the model
R1	Receiver in the source room (Room 1)
R4	Governing receiver in the receiving room (Room 2)
Al	Aluminium
mm	Millimetre
m ²	Square metre

1 Introduction

1.1 Background and Motivation

Acoustic privacy is a defining quality attribute of hotel accommodation. Noise transmitted from adjacent rooms can disturb sleep, reduce comfort and compromise the privacy expected by hotel guests [1]. The separating wall is a key element governing airborne sound transfer between adjacent guest rooms. Its construction strongly influences the transmitted sound that reaches the neighbouring room, although flanking paths and junction details can also affect real-building performance.

In contemporary interior schemes, designers increasingly specify metal-finished surfaces — brushed or anodised aluminium panels and metal-faced partition systems — for their durability, hygiene and clean visual character. Metal finishes, however, raise an immediate acoustic question. A thin metal sheet is lightweight and stiff, and on its own provides little resistance to sound; yet the same metal aesthetic can be retained while building a far more capable assembly behind it. The motivation for this study is to quantify, by simulation, how much the acoustic separation between two adjacent hotel rooms depends on the way a metal-finish separating wall is constructed.

The room-acoustics program ODEON is used to carry out this evaluation. ODEON applies geometrical-acoustics methods to predict how sound propagates within and between modelled spaces, and it allows the acoustic behaviour of each surface — including the transmission characteristics of a separating wall — to be defined explicitly [2], [3]. This makes it well suited to a controlled comparison in which only the separating wall is changed.

1.2 Aim and Objectives

The aim of this report is to compare three metal-finish separating-wall assemblies and to identify, on engineering grounds, the assembly that best protects acoustic privacy between two adjacent hotel guest rooms. The specific objectives are:

- To build a single, consistent two-room ODEON model representing two adjoining guest rooms with a fixed sound source and defined receiver positions.
- To define three separating-wall assemblies that share a metal architectural finish but differ in construction, and to assign each a transmission-loss schedule.
- To extract the A-weighted sound-pressure level $SPL(A)$ and the octave-band spectrum at the source-room receiver (R1) and the receiving-room receiver (R4), and to compute the A-weighted sound-level reduction ΔL_A for each case.

- To analyse the differences between the three cases across the frequency range, including any frequency-dependent behaviour, and to interpret them in terms of mass, cavity and damping effects.
- To recommend the most suitable assembly and to state clearly the assumptions and limitations of the comparison.

1.3 Scope and Adaptation of the Original Brief

The original teaching exercise on which this work is based asked for a comparison of separating walls of different thicknesses. In this report that brief has been deliberately adapted into a comparison of three distinct wall build-ups rather than three thicknesses of a single material. This adaptation is an intentional and justified engineering decision, not a deviation in error: in real hotel design the practical question facing an engineer is rarely "how thick should one material be?" but rather "which wall system should be specified?". Comparing genuinely different assemblies — a bare sheet, a sandwich panel and an insulated stud partition — exposes the roles of mass, cavity separation and absorptive infill in a way that simply increasing the thickness of one material cannot. The same controlled-comparison discipline is retained: only the separating wall changes between cases, and all results are reported on a like-for-like basis. The scope is limited to airborne sound transmission between the two modelled rooms.

2 Project Background and Acoustic Problem

2.1 Building and Room Configuration

The study concerns two adjoining guest rooms in a hotel located in Dammam, in the Eastern Province of Saudi Arabia. Both rooms share a common separating wall, and it is the airborne sound passing from one room to the other across this wall that the model evaluates. The principal project parameters are summarised in Table 1. The geometry, the source and the non-tested surfaces are identical in every simulation; only the separating-wall assembly is varied, so that any change in the predicted levels can be attributed solely to the wall.

Table 1. *Principal project and model parameters held constant across all three cases.*

Parameter	Value
Project location	Dammam, Eastern Province, Saudi Arabia
Building type	Hotel — two adjacent guest rooms
Phenomenon analysed	Airborne sound transmission between rooms
Simulation tool	ODEON (geometrical room acoustics)
Guest-room floor area	32.4 m ²
Floor-to-floor height	4.20 m

Parameter	Value
Net clear room height	2.90 m
Ceiling void depth	1.00 m
Source-room level, SPL(A)	58.9 dB(A) (identical for all cases)
Separating-wall variants	Three assemblies (Case 1, Case 2, Case 3)

2.2 Source and Receiver Layout

A single omnidirectional sound source, labelled P1, is placed in the first room to represent a disturbing noise such as speech or television. Two receivers govern the analysis: R1, located in the source room, records the reference level on the noisy side of the wall, while R4, located in the receiving room, records the sound transmitted from the source into that room — the source-related contribution that reaches the neighbour through the separating wall, rather than the total ambient noise of an occupied room. Additional receiver positions present in the model are not used in the headline comparison; concentrating on the R1–R4 pair gives a clear and repeatable measure of how much the wall reduces the level from one room to the next. The source and receiver layout is listed in Table 2 and shown later in the annotated model (Figure 2).

Table 2. Sound source and receiver positions used in the simulations.

Position	Location	Coordinates (x, y, z), m	Role in analysis
P1	Room 1 (source)	Near room centre	Omnidirectional disturbing source
R1	Room 1 (source)	(3.29, 6.10, 0.75)	Reference level in the source room
R4	Room 2 (receiving)	(5.58, 6.10, 0.75)	Governing receiver; transmitted source contribution
R2, R3, R5, R6	Both rooms	Additional positions	Present in model; outside headline comparison

2.3 The Airborne-Sound Privacy Problem

When the source operates in Room 1, sound energy builds up in that room and bears on the separating wall. Part of that energy is transmitted through the wall, re-radiated into Room 2, and heard by the neighbour. The level reaching Room 2 depends primarily on how effectively the wall resists transmission across the audible frequency range. Speech and television noise carry significant energy in the mid frequencies, around 500 Hz to 2 kHz, where the human ear is also most sensitive; a wall that performs poorly in this region will allow intelligible, intrusive sound to pass even if its overall mass is modest. A wall that performs well across the mid and high frequencies, by contrast, reduces the neighbour's level to a point where any residual sound is faint and unintelligible. The objective of the separating wall is therefore not merely to lower

the overall level but to control the frequencies that matter for privacy. The three assemblies examined in this report represent three very different answers to that objective.

3 Theoretical Background

3.1 Sound-Pressure Level (SPL)

Sound-pressure level expresses an acoustic pressure on a logarithmic scale relative to a fixed reference pressure of 20 micropascals, the nominal threshold of human hearing. It is reported in decibels (dB). Because the scale is logarithmic, equal numerical steps correspond to equal ratios of acoustic power rather than equal increments of pressure: a reduction of 10 dB corresponds to a tenfold decrease in sound intensity and is perceived, very approximately, as a halving of loudness [4]. Sound-pressure level is the basic quantity that ODEON predicts at each receiver, both as a single overall figure and band by band across the spectrum.

3.2 A-Weighted Sound-Pressure Level, SPL(A)

The human ear does not respond equally to all frequencies; it is far less sensitive to low-frequency sound than to mid-frequency sound at the same physical level. The A-weighting network applies a frequency-dependent correction that mirrors this response, attenuating the low and very high frequencies before the band levels are combined into a single figure. The result, the A-weighted sound-pressure level SPL(A), expressed in dB(A), correlates well with the loudness and the disturbance of everyday noise such as speech and television [5]. For a hotel privacy assessment, SPL(A) is the most meaningful single descriptor of what a guest actually experiences, and it is adopted in this report as the headline level at each receiver.

3.3 Transmission Loss and the Sound-Reduction Index

The ability of a building element to resist the passage of airborne sound is described by its transmission loss (TL), also called the sound-reduction index (R) and expressed in decibels as a function of frequency. A higher value means that more sound energy is stopped by the element. Transmission loss is governed chiefly by the so-called mass law, under which the loss rises with both the surface mass of the element and the frequency of the sound, so that heavier walls and higher frequencies are attenuated more strongly [6]. Real assemblies depart from the simple mass law in important ways: double-leaf constructions separated by a cavity can achieve far more than their combined mass would suggest, while resonances and coincidence effects can produce local dips in performance at particular frequencies. In ODEON, the transmission-loss schedule of the separating wall is supplied as an input that defines, band by band, how

much the wall attenuates sound passing through it; it is the property that distinguishes the three cases in this study.

3.4 Selection of the Main Decision Metric

Because the study concerns sound transmitted from one room to another, the governing quantity is the difference in A-weighted level between the source room and the receiving room rather than any property of a single room in isolation. The main decision metric adopted here is the A-weighted sound-level reduction between the rooms, ΔL_A , together with the absolute A-weighted level at the receiving-room receiver R4. This keeps the analysis focused on airborne transmission between the two rooms and on the quantities most relevant to acoustic privacy.

3.5 A-Weighted Sound-Level Reduction (ΔL_A)

The A-weighted sound-level reduction is defined as the difference between the A-weighted sound-pressure level at the source-room receiver R1 and that at the receiving-room receiver R4:

$$\Delta L_A = \text{SPL}(A)_{R1} - \text{SPL}(A)_{R4}$$

A larger ΔL_A indicates a greater drop in level across the separating wall and therefore better acoustic separation, while a higher $\text{SPL}(A)$ at R4 — the predicted source-related transmitted level in the receiving room — indicates a louder, less private result. Throughout this report the source-room level is essentially constant, so ΔL_A and the R4 level move together: the assembly that produces the lowest R4 level also produces the largest reduction. These two quantities, read jointly, form the basis of the comparison and of the final recommendation.

4 Methodology

4.1 Rationale for the Wall-System Comparison

As set out in Section 1.3, the original brief to compare wall thicknesses has been adapted into a comparison of three different wall systems. The methodological reason for this choice is that the three dominant mechanisms of airborne sound insulation — added mass, separation of leaves by a cavity, and absorption within that cavity — cannot all be exercised by changing the thickness of a single material. A bare metal sheet isolates the effect of mass alone; a sandwich panel adds an absorptive core between two metal skins; and an insulated stud partition introduces a genuine double-leaf system with a damped cavity. By holding the room, the source and every non-tested surface constant and varying only the separating wall, the simulation isolates the contribution of the wall system itself, which is precisely the variable an engineer

controls when specifying a hotel partition. The adaptation therefore strengthens, rather than weakens, the engineering value of the comparison.

4.2 The Two-Room ODEON Model

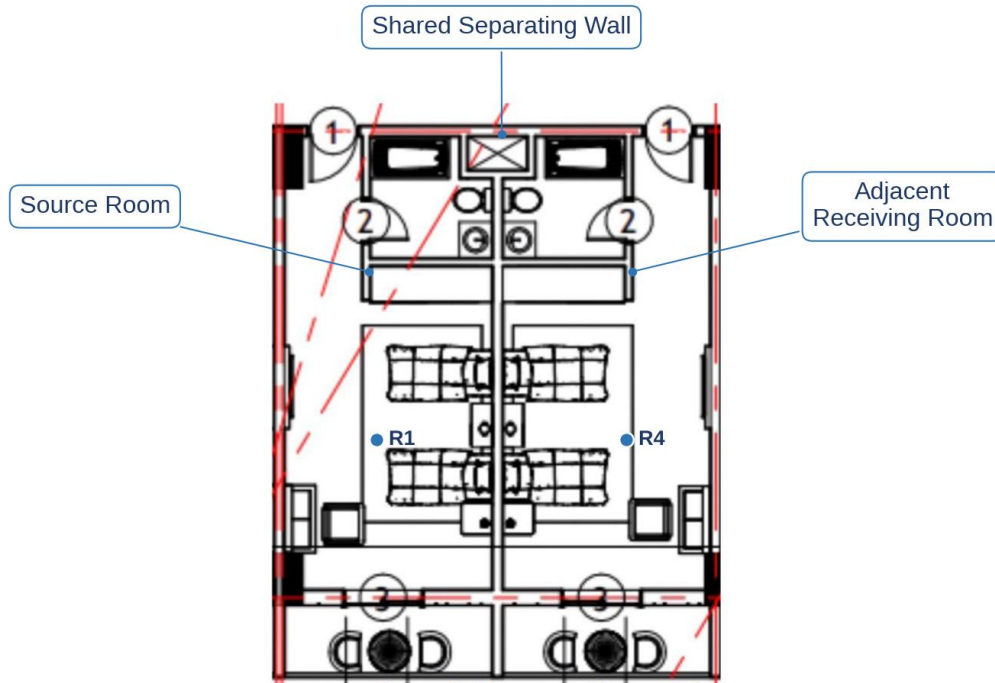


Figure 1. Architectural plan of the two adjacent hotel guest rooms used in the ODEON simulation, showing the shared separating wall between the source room and the adjacent receiving room.

The architectural plan illustrates the two adjacent hotel guest rooms selected for the acoustic-transmission study. The rooms share a separating wall, which is the only construction element varied between the three simulation cases. The left room was used as the source room, while the adjacent room was used as the receiving room. This controlled configuration allows the effect of each metal-based wall assembly on transmitted sound to be evaluated consistently.

A single ODEON model of two adjacent guest rooms was constructed and used for all three cases. The source room (Room 1) contains the source P1 and the reference receiver R1; the receiving room (Room 2) contains the governing receiver R4. The separating wall lies between the two rooms, and it is this surface whose transmission-loss schedule is exchanged from case to case. Figure 2 shows the model with the source, the two analysed receivers and the tested separating wall annotated. Keeping a single geometry guarantees that differences in the predicted levels arise only from the wall and not from any change in room shape, source position or receiver location.

ROOM 1 — SOURCE ROOM

ROOM 2 — RECEIVING ROOM

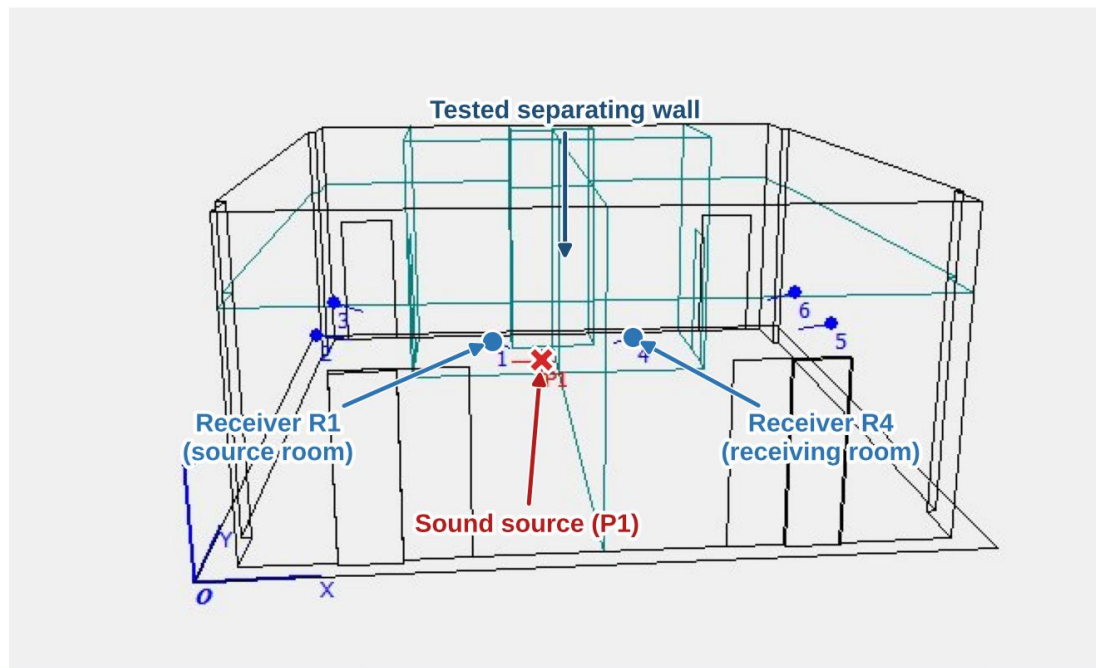


Figure 2. Annotated two-room ODEON model identifying the sound source *P1*, the source-room receiver *R1*, the receiving-room receiver *R4*, and the tested separating wall.

4.3 Simulation Workflow

The same seven-stage workflow, summarised in Figure 3, was followed for each of the three cases:

1. Build the two adjacent guest-room geometry in ODEON, including the separating wall and all bounding surfaces.
2. Place the omnidirectional source *P1* in Room 1 and define the receivers *R1* (source room) and *R4* (receiving room).
3. Assign the separating-wall transmission-loss schedule for the case under consideration, leaving all other surface properties unchanged.
4. Run the ODEON airborne-transmission simulation for the configured case.
5. Extract the A-weighted sound-pressure level $SPL(A)$ at *R1* and *R4* together with the octave-band SPL spectrum at each receiver.
6. Compute the A-weighted sound-level reduction $\Delta L_A = SPL(A)_{R1} - SPL(A)_{R4}$ for the case.
7. Compare the three cases across the overall and band-by-band results and form an engineering recommendation.

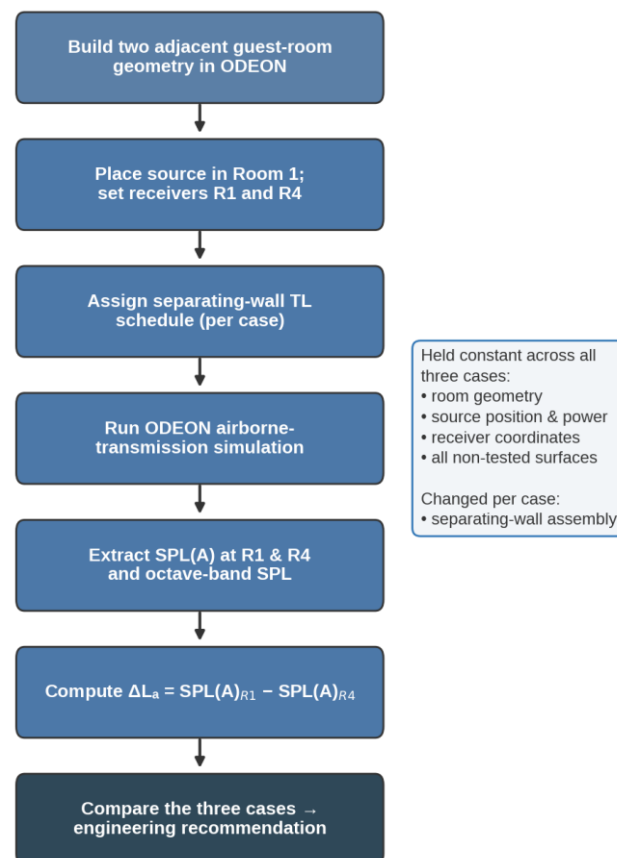


Figure 3. Seven-stage simulation workflow applied identically to each case, with the parameters held constant and the single parameter varied between cases.

4.4 Controlled and Varied Parameters

The integrity of the comparison rests on a strict separation between what is held constant and what is varied. Held constant across all three cases are the room geometry and dimensions, the position and acoustic power of the source, the coordinates of every receiver, and the acoustic properties of all surfaces other than the separating wall. Varied between cases is one parameter only: the separating-wall assembly, represented in the model by its transmission-loss schedule. Because a single quantity changes, every difference in the predicted SPL(A) and octave-band levels can be attributed unambiguously to the wall system, and the three cases can be ranked on a fully like-for-like basis.

5 Wall Assemblies and Transmission-Loss Inputs

Three separating-wall assemblies were defined, all presenting an aluminium architectural finish to both rooms but differing fundamentally in construction. Table 3 summarises the three build-ups; the sections that follow describe each in turn and give its transmission-loss schedule. The transmission-loss values used in this study are engineering estimates assigned for the purpose

of comparison: they are model inputs, not laboratory-certified data, and each is identified as such below.

Table 3. Construction of the three metal-finish separating-wall assemblies compared in the study.

Case	Build-up (source → receiving room)	Total, mm	Finish	Acoustic role
Case 1	Single aluminium sheet, 1.0 mm	1.0	Aluminium	Lightweight baseline (mass only)
Case 2	1.5 mm Al / 100 mm rockwool (assumed) / 1.5 mm Al	103	Aluminium	Mass with absorptive core
Case 3	1.0 mm Al / 12.5 mm gypsum / 50 mm metal-stud cavity with rockwool / 12.5 mm gypsum / 1.0 mm Al	77	Aluminium	Double-leaf insulated stud partition

5.1 Case 1 — Single Aluminium Sheet

Case 1 is the lightweight baseline: a single 1.0 mm aluminium sheet forming the entire separating wall, with the metal surface exposed to both rooms. The cross-section is shown in Figure 4. With so little surface mass, the assembly is expected to provide only modest attenuation, governed essentially by the mass law and therefore weakest at low frequencies and improving gradually towards the high frequencies. The transmission-loss schedule assigned to Case 1, listed in Table 4, is an estimate based on mass-law behaviour for a thin aluminium sheet; the values are zero in the lowest bands, reflecting the negligible insulation of a single light skin there, and rise steadily with frequency.

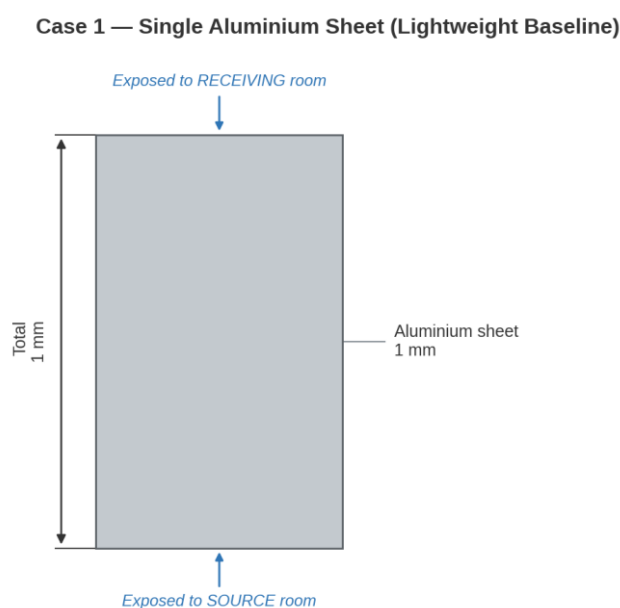


Figure 4. Case 1 — single 1.0 mm aluminium sheet (lightweight baseline), shown to scale with the faces exposed to each room.

Table 4. *Estimated one-third-octave transmission-loss schedule assigned to Case 1 (mass-law estimate for a thin aluminium sheet).*

Freq, Hz	TL, dB	Freq, Hz	TL, dB
50	0.00	800	19.69
63	0.00	1000	21.63
80	0.00	1250	23.57
100	1.63	1600	25.71
125	3.57	2000	27.65
160	5.71	2500	29.59
200	7.65	3150	31.59
250	9.59	4000	33.67
315	11.59	5000	35.61
400	13.67	6300	37.61
500	15.61	8000	39.69
630	17.61	10000	41.63

5.2 Case 2 — Aluminium–Rockwool–Aluminium Sandwich

Case 2 retains the metal aesthetic but builds a sandwich panel: two 1.5 mm aluminium skins enclosing a 100 mm rockwool core, giving a total thickness of 103 mm. The cross-section is shown in Figure 5. The two skins add mass and the absorptive core damps sound within the panel, so the assembly should outperform the bare sheet across most of the spectrum and show markedly better low-frequency behaviour. The transmission-loss schedule for Case 2 is given in Table 5. The 100 mm rockwool core is a preliminary assumed value adopted for this comparison; the schedule shows the higher low-frequency loss expected of a heavier, damped panel, together with the characteristic irregularities of a real multi-layer construction rather than a smooth mass-law line.

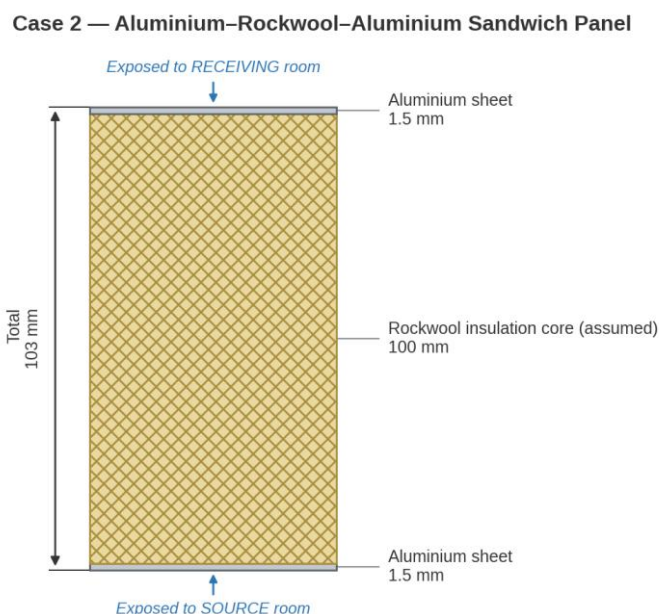


Figure 5. Case 2 — aluminium–rockwool–aluminium sandwich panel (103 mm total), shown to scale.

Table 5. Estimated one-third-octave transmission-loss schedule assigned to Case 2 (preliminary, assuming a 100 mm rockwool core).

Freq, Hz	TL, dB	Freq, Hz	TL, dB
50	30.20	800	31.80
63	24.80	1000	32.30
80	25.10	1250	30.10
100	16.50	1600	25.50
125	23.50	2000	27.10
160	24.00	2500	34.00
200	25.60	3150	37.70
250	24.60	4000	39.80
315	26.90	5000	43.70
400	28.00	6300	46.50
500	29.30	8000	49.00
630	30.60	10000	51.00

5.3 Case 3 — Metal-Finish Insulated Stud Partition

Case 3 is a genuine double-leaf partition finished in metal: from each room inward, a 1.0 mm aluminium finish, a 12.5 mm gypsum board, a 50 mm metal-stud cavity filled with rockwool, a second 12.5 mm gypsum board and a second 1.0 mm aluminium finish, for a total thickness of 77 mm. The cross-section is shown in Figure 6. This arrangement combines mass in the gypsum boards, separation of the two leaves by the stud cavity and absorption by the rockwool infill — the three mechanisms of high-performance airborne insulation acting together. Its transmission-loss schedule, listed in Table 6, is a conservative engineering estimate; it rises

smoothly and reaches the highest values of the three cases across almost the entire range, consistent with an effective double-leaf system.

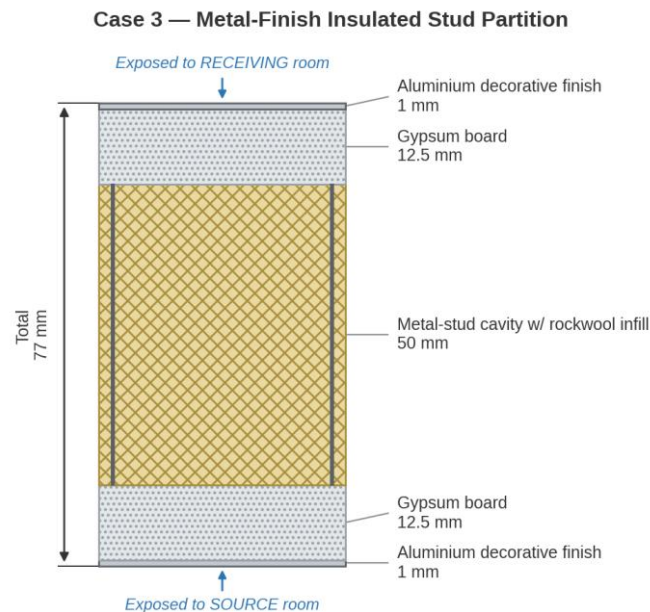


Figure 6. Case 3 — metal-finish insulated stud partition with a rock-wool-filled cavity (77 mm total), shown to scale.

Table 6. Estimated one-third-octave transmission-loss schedule assigned to Case 3 (conservative engineering estimate for a double-leaf insulated stud partition).

Freq, Hz	TL, dB	Freq, Hz	TL, dB
50	17.00	800	43.00
63	18.00	1000	45.00
80	20.00	1250	46.00
100	22.00	1600	47.00
125	24.00	2000	49.00
160	27.00	2500	50.00
200	29.00	3150	51.00
250	32.00	4000	53.00
315	34.00	5000	54.00
400	37.00	6300	55.00
500	39.00	8000	56.00
630	41.00	10000	57.00

5.4 Comparison of the Transmission-Loss Schedules

Plotting the three schedules together, in Figure 7, makes their relationship clear. Case 1 is the lowest curve throughout, with no insulation in the lowest bands and only moderate values even at high frequency. Case 3 is the highest curve across almost the entire range and rises smoothly, reflecting an effective double-leaf system. Case 2 sits between the two for much of the spectrum

but behaves irregularly: at the very lowest bands its assumed mass gives it a higher loss than Case 3, while around the middle of the range a dip brings it close to, and at one or two bands below, the Case 1 line. These input differences are the direct cause of the differences in the simulated room-to-room results presented in Section 6.

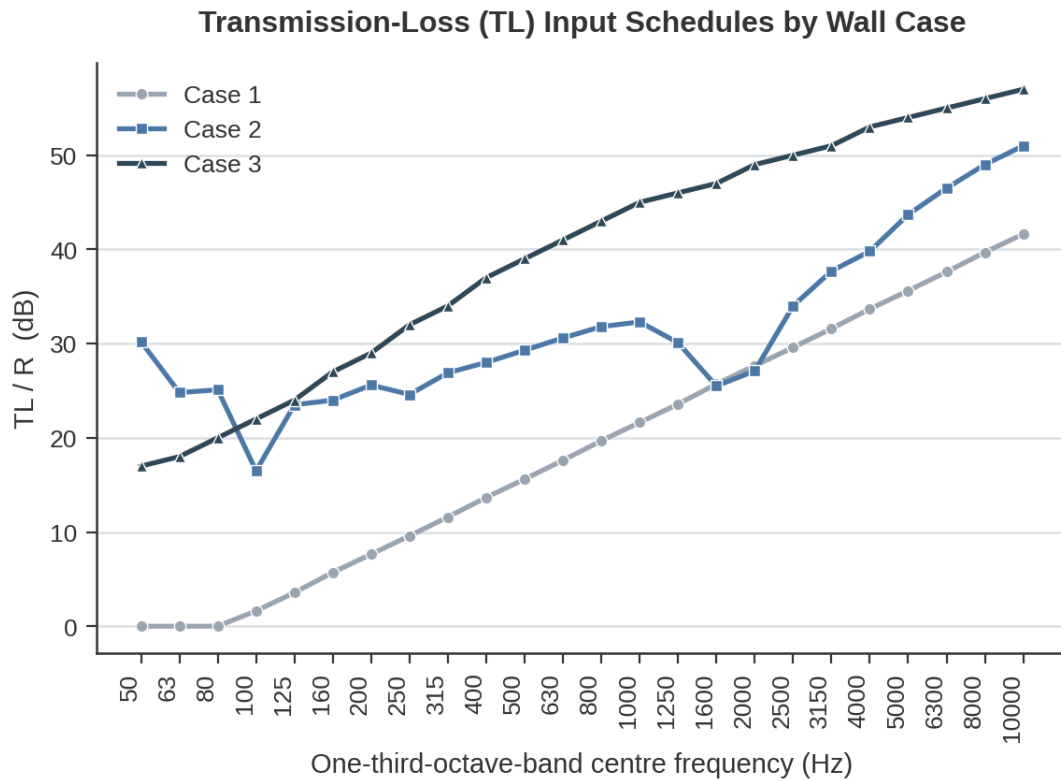


Figure 7. Transmission-loss input schedules for the three cases across the one-third-octave bands.

6 Simulation Results

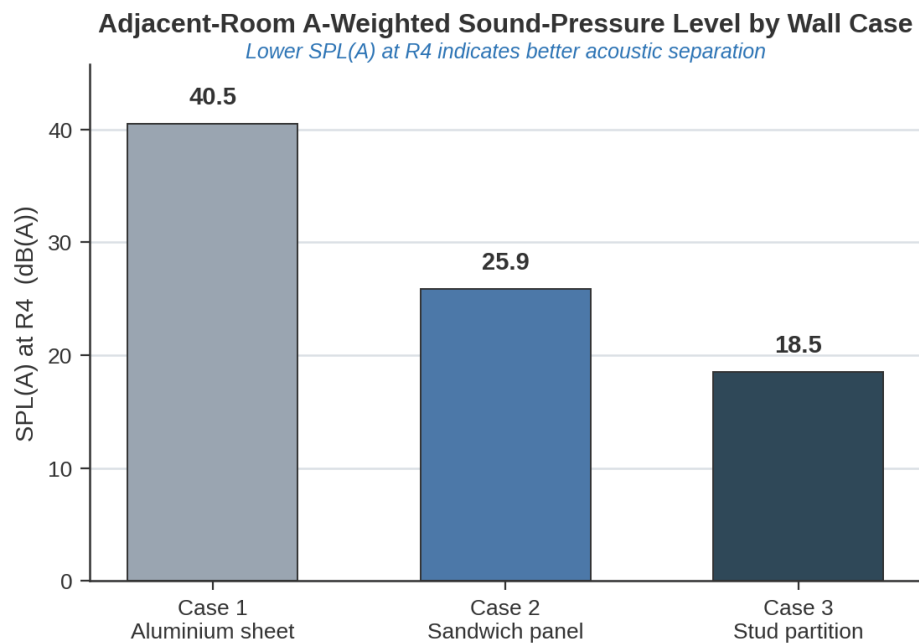
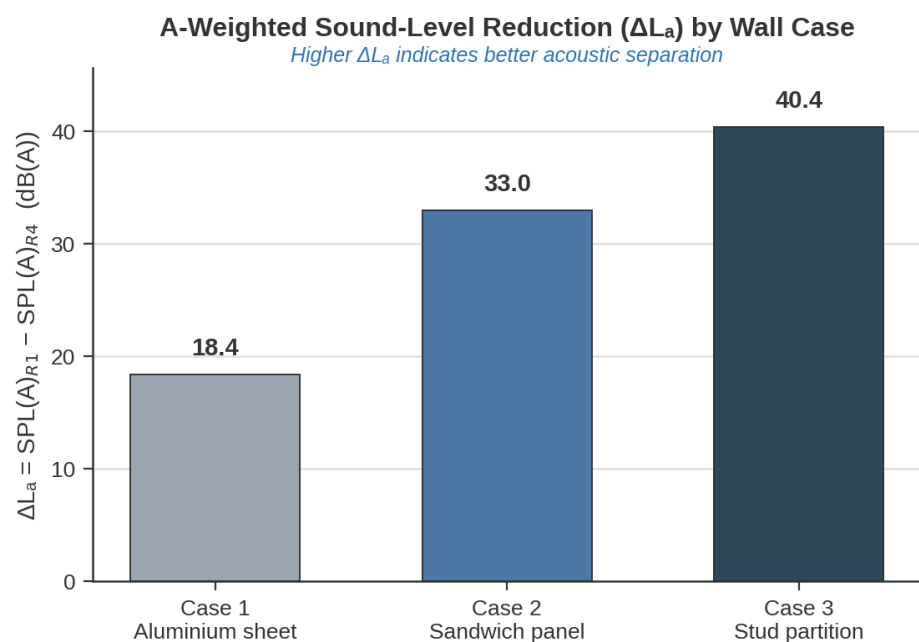
This section reports the simulated results for the three cases without interpretation; the engineering discussion follows in Section 7. In every case the source-room level at R1 corresponds to an A-weighted level of 58.9 dB(A), so the receiving-room level at R4 and the reduction between the rooms are determined entirely by the separating wall.

6.1 A-Weighted Levels and Overall Reduction

Table 7 summarises the headline results. The predicted transmitted level at the receiving-room receiver R4 falls from 40.5 dB(A) in Case 1 to 25.9 dB(A) in Case 2 and 18.5 dB(A) in Case 3, while the A-weighted sound-level reduction ΔL_A rises correspondingly from 18.4 to 33.0 and 40.4 dB(A). Relative to the Case 1 baseline, Case 2 improves the reduction by 14.6 dB and Case 3 by 22.0 dB; Case 3 improves on Case 2 by a further 7.4 dB. Figure 8 shows the receiving-room levels and Figure 9 the reductions.

Table 7. Summary of A-weighted results: source-room level, receiving-room level, reduction and improvement relative to Case 1.

Case	Assembly	Source SPL(A), dB(A)	R4 SPL(A), dB(A)	Reduction ΔL_a , dB(A)	vs Case 1, dB
Case 1	Aluminium sheet	58.9	40.5	18.4	—
Case 2	Sandwich panel	58.9	25.9	33.0	+14.6
Case 3	Stud partition	58.9	18.5	40.4	+22.0

**Figure 8.** Receiving-room A-weighted sound-pressure level $SPL(A)$ at R4 for each case (lower is better).**Figure 9.** A-weighted sound-level reduction ΔL_a between the rooms for each case (higher is better).

6.2 Octave-Band Levels at the Receiver

The single A-weighted figure conceals important frequency detail. Table 8 lists the octave-band sound-pressure levels at R4 for the three cases, and Figure 10 plots the same spectra. Reading across the bands, Case 3 gives the lowest receiving-room level in every band from 125 Hz upward and falls steeply at high frequency, whereas Case 1 remains high across the low and mid bands. Case 2 lies between the two over most of the range. The one exception occurs at 63 Hz, where the Case 2 level (25.6 dB) is below the Case 3 level (33.6 dB); this low-frequency crossover is examined in Section 7.3. The negative band levels at the highest frequencies simply indicate levels far below the reference and are acoustically negligible.

Table 8. Octave-band sound-pressure level at the receiving-room receiver R4, by case (dB).

Octave band, Hz	Case 1	Case 2	Case 3
63	51.9	25.6	33.6
125	49.5	32.9	28.9
250	44.4	28.1	22.5
500	39.8	25.9	16.4
1000	26.3	16.4	3.2
2000	13.4	13.1	-7.7
4000	2.1	-4.3	-17.0
8000	-5.0	-14.1	-21.5

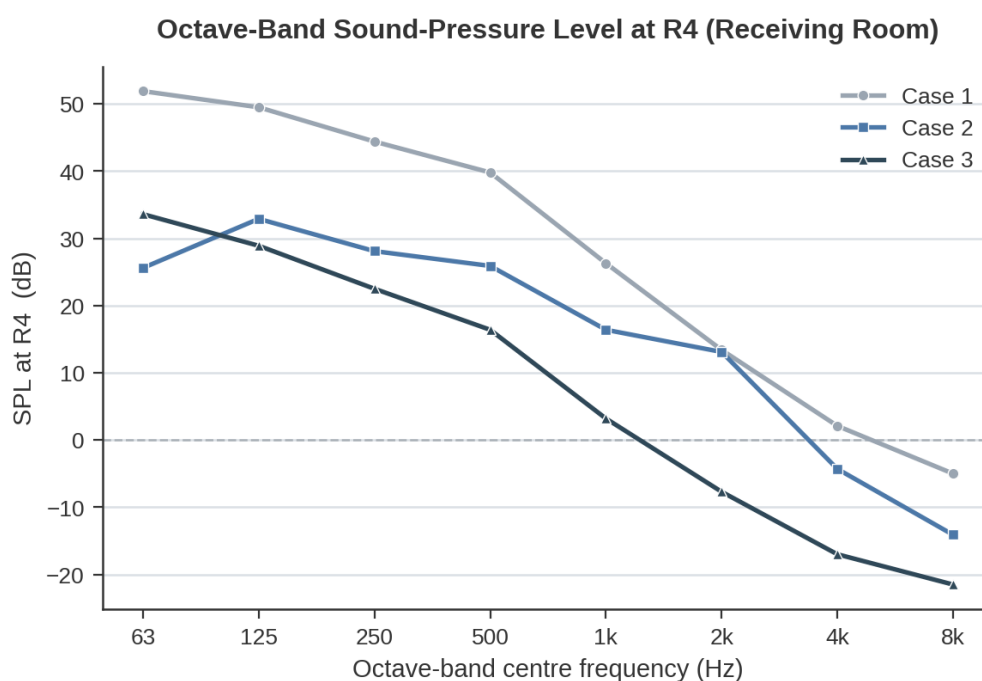


Figure 10. Octave-band sound-pressure level at R4 for the three cases, showing the 63 Hz crossover between Case 2 and Case 3.

7 Comparative Analysis and Discussion

7.1 Octave-Band Sound-Level Reduction

The clearest picture of wall performance comes from the reduction achieved in each octave band, obtained as the difference between the source-room and receiving-room levels ($R1 - R4$). These band reductions are listed in Table 9 and plotted in Figure 11. Case 3 achieves the greatest reduction in every band from 125 Hz upward, exceeding 40 dB by 500 Hz and rising above 60 dB at the top of the range. Case 2 provides a strong reduction at the lowest band and good reductions across the upper range, but its curve dips in the lower-middle frequencies before recovering. Case 1 is the weakest throughout, beginning at only a few decibels in the lowest bands. The ranking by band reduction is consistent with the receiving-room spectra of Section 6.2.

Table 9. Octave-band sound-level reduction ($R1 - R4$) for the three cases (dB).

Octave band, Hz	Case 1	Case 2	Case 3
63	2.9	28.6	20.6
125	5.3	21.8	25.8
250	12.0	28.3	33.9
500	20.3	34.2	43.7
1000	25.6	35.5	48.7
2000	32.2	32.5	53.3
4000	39.2	45.6	58.3
8000	46.0	55.1	62.5

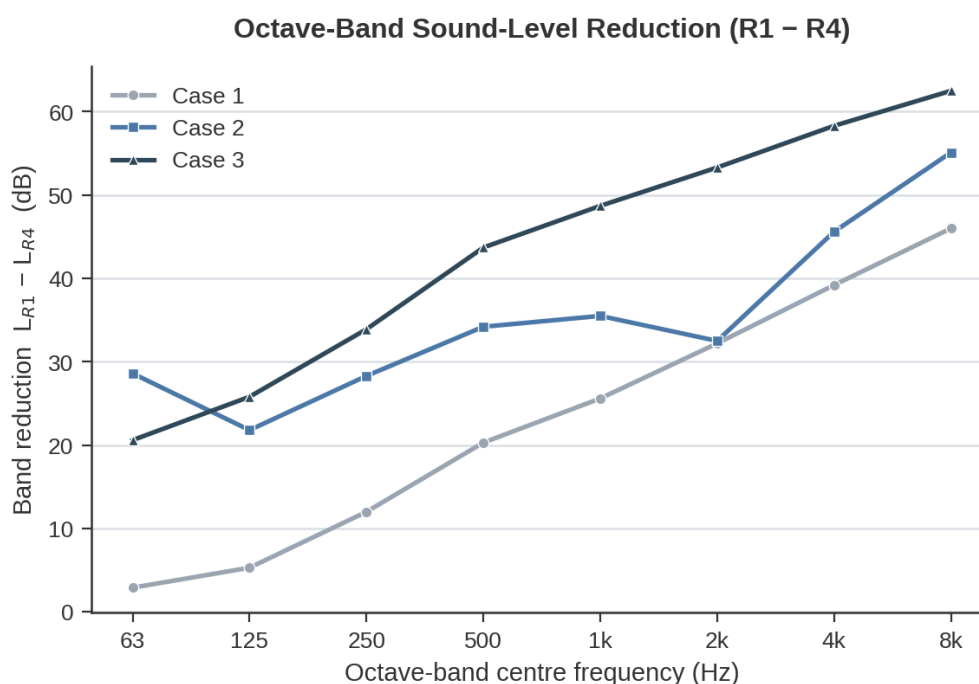


Figure 11. Octave-band sound-level reduction (R1 – R4) for the three cases across the audible range.

7.2 Case-by-Case Engineering Interpretation

Case 1 behaves exactly as a single lightweight skin should. With only the mass of a 1 mm aluminium sheet to resist transmission, it offers almost no reduction at low frequency and only a gradual improvement towards the high frequencies, following the mass law. The predicted transmitted level at R4 of 40.5 dB(A) is high enough that speech and television from the neighbouring room would remain clearly audible, and the assembly is therefore unsuitable as a guest-room separating wall on acoustic grounds.

Case 2 demonstrates the benefit of adding mass and an absorptive core. The two aluminium skins and the assumed rockwool fill raise the reduction substantially over the baseline, lowering the receiving-room level to 25.9 dB(A) and increasing the reduction to 33.0 dB(A). Its performance is strong at the extremes of the range but uneven through the lower-middle frequencies, where the input schedule dips; this irregularity is typical of a real multi-layer panel and prevents Case 2 from matching the smoothness of an optimised double-leaf system.

Case 3 gives the most complete performance. By combining the mass of two gypsum boards, the separation of a stud cavity and the absorption of a rockwool infill, it produces the lowest receiving-room level, 18.5 dB(A), and the largest reduction, 40.4 dB(A), while remaining thinner overall than the Case 2 sandwich. Its band reductions are the highest from 125 Hz upward and increase steadily with frequency, which is the desirable behaviour for controlling

speech and television noise. Among the three, Case 3 is the assembly that an acoustic engineer would expect to deliver dependable inter-room privacy.

7.3 Low-Frequency Behaviour and the 63 Hz Crossover

One result runs against the overall ranking and is reported here transparently rather than smoothed over. At the 63 Hz octave band, Case 2 outperforms Case 3: the receiving-room level is 25.6 dB in Case 2 against 33.6 dB in Case 3 (Table 8), and the band reduction is 28.6 dB against 20.6 dB (Table 9). At this single low-frequency band the sandwich panel attenuates more than the stud partition.

This behaviour is consistent with established double-leaf theory. A double-leaf partition such as Case 3 has a mass–air–mass resonance in the low-frequency region, governed by the mass of its leaves and the depth and stiffness of the cavity between them. Below and around that resonance the two leaves tend to move together and the cavity provides little benefit, so the partition can perform worse there than a heavier, more massive single element. The relatively shallow 50 mm cavity of Case 3 places this low-frequency weakness within the bottom of the audible range, while the heavier, mass-dominated Case 2 sandwich retains a higher loss at the very lowest band. The exact resonance frequency depends on the detailed construction and has not been determined here; the point is simply that the 63 Hz band lies in the region where double-leaf behaviour is least favourable.

The practical significance of this crossover is small. Speech, television and most everyday disturbance carry comparatively little energy at 63 Hz, the ear is relatively insensitive there, and the A-weighting network applied in computing SPL(A) attenuates this band heavily. Across the mid and high frequencies that dominate privacy, Case 3 is decisively superior, and its overall ΔL_A of 40.4 dB(A) exceeds that of Case 2 by 7.4 dB. The single-band advantage of Case 2 at 63 Hz therefore does not alter the overall conclusion, although it is a genuine characteristic of the assemblies and would matter more for a source rich in low-frequency content.

8 Engineering Recommendation

8.1 Decision Criteria

The selection of a separating-wall assembly was made against five criteria that together capture acoustic performance, design fit and buildability: the resulting adjacent-room A-weighted level; the overall A-weighted reduction between the rooms; control of the mid- and high-frequency range most important for privacy; the match to the intended metal-finish design; and the realism

of the construction for a hotel project. Table 10 sets out a qualitative assessment of the three cases against these criteria.

Table 10. *Qualitative decision matrix for the three separating-wall assemblies.*

Criterion	Case 1	Case 2	Case 3
Adjacent-room SPL(A)	Weak	Good	Best
Overall reduction ΔL_A	Weak	Good	Best
Mid/high-frequency control	Weak	Good	Best
Match to metal-finish design	Partial	Good	Best
Construction realism	Low	Moderate	High
Final recommendation	Not selected	Acceptable alternative	Recommended

8.2 Comparative Assessment

Figure 12 brings the two governing quantities together. On reduction, the cases rank Case 3 (40.4 dB(A)) above Case 2 (33.0 dB(A)) above Case 1 (18.4 dB(A)); on receiving-room level the order is the mirror image, with Case 3 lowest at 18.5 dB(A). The two views agree completely, because the source-room level is constant: the assembly with the largest ΔL_A is necessarily the one that leaves the quietest receiving room. Case 3 leads on both measures by a clear margin, Case 2 is a solid middle option, and Case 1 trails well behind.

Comparative decision summary — Case 3 recommended

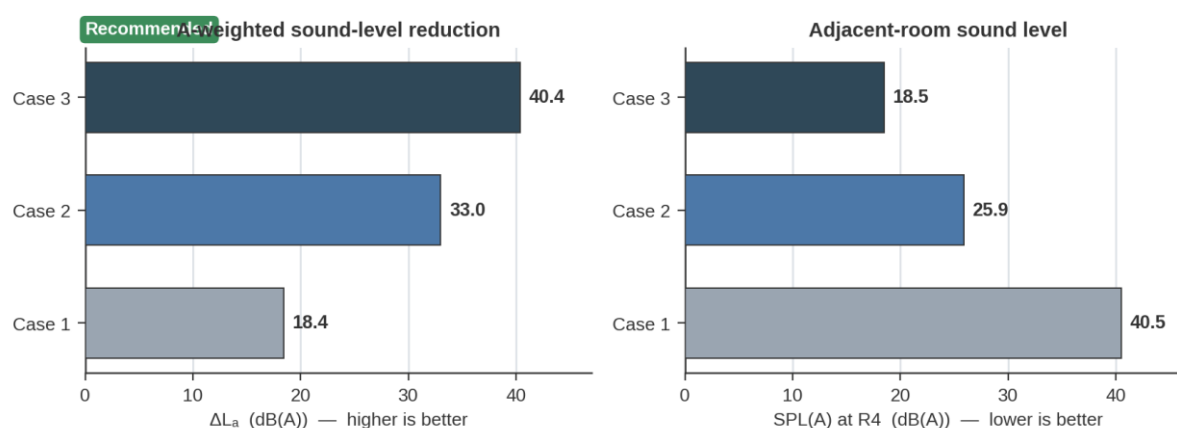


Figure 12. *Comparative decision summary: A-weighted reduction ΔL_A and adjacent-room level SPL(A) for the three cases; Case 3 is recommended.*

8.3 Recommended Wall System and Limitations

On the balance of the criteria, Case 3 — the metal-finish insulated stud partition — is recommended as the separating wall between the two adjacent guest rooms. It delivers the lowest receiving-room level and the greatest overall reduction, the strongest control of the speech-critical mid and high frequencies, and a faithful metal architectural finish, all within a

77 mm build-up that is thinner than the Case 2 sandwich and entirely realistic to construct with standard metal-stud, gypsum and rockwool components. Case 2 is recorded as an acceptable alternative where a sandwich-panel construction is preferred, since it still provides a substantial improvement over the bare sheet, while Case 1 is not suitable as a guest-room separating wall on acoustic grounds.

This recommendation is made within the limits of the study and should be read alongside them. The transmission-loss schedules used for all three cases are engineering estimates assigned as model inputs, not laboratory-certified data, and the Case 2 core thickness in particular is a preliminary assumption. The receiving-room levels reported at R4 represent the sound transmitted from the source through the separating wall, not the total background noise that would be present in an occupied hotel room with its own internal and external sources. The analysis evaluates the modelled configuration and the chosen receiver positions, and it considers airborne transmission through the separating wall; flanking transmission along adjoining structures and the effects of workmanship and detailing at junctions are not represented. Before construction, the selected assembly should be confirmed against manufacturer test data and the relevant sound-insulation standards [7], [8], and verified on site where required [9]. More broadly, national building regulations commonly set minimum sound-insulation requirements between dwellings, and any final design would be assessed against the applicable local code [10]. Within these bounds, the comparison provides a sound and consistent basis for preferring Case 3.

9 Conclusion

This report used the ODEON room-acoustics program to compare three metal-finish separating-wall assemblies between two adjacent hotel guest rooms in Dammam. A single two-room model with a fixed source and defined receivers in the source room (R1) and the receiving room (R4) was used throughout, and only the separating wall was varied, so that all differences in the predicted levels could be attributed to the wall system alone.

From an identical source-room level of 58.9 dB(A), the three assemblies produced predicted transmitted levels at R4 of 40.5, 25.9 and 18.5 dB(A) and A-weighted reductions ΔL_A of 18.4, 33.0 and 40.4 dB(A) for Cases 1, 2 and 3 respectively. Case 1, the single aluminium sheet, behaved as a lightweight mass-law element and offered inadequate privacy. Case 2, the aluminium–rockwool–aluminium sandwich, improved the reduction by 14.6 dB over the baseline but performed unevenly through the lower-middle frequencies. Case 3, the metal-finish

insulated stud partition, combined mass, cavity separation and absorption to give the lowest receiving-room level and the largest reduction, with the best control of the mid and high frequencies that matter most for speech privacy. The one departure from this ranking, a single-band crossover at 63 Hz where Case 2 attenuates more than Case 3, was explained by the low-frequency mass–air–mass behaviour of double-leaf partitions and shown to be of little practical consequence after A-weighting.

Case 3 is therefore recommended as the separating wall, with Case 2 as an acceptable alternative. The conclusion rests on transmission-loss inputs that are engineering estimates rather than certified data, and the recommended assembly should be confirmed against manufacturer testing and the applicable standards before construction. Within these limits, the study gives a clear, consistent and defensible basis for the choice.

10 References

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- [10] B. Rasmussen, "Sound insulation between dwellings — Requirements in building regulations in Europe," *Applied Acoustics*, vol. 71, no. 4, pp. 373–385, 2010.

Appendix A Consolidated Transmission-Loss Schedules

Table 11 lists, in full, the one-third-octave transmission-loss schedules assigned to the three cases as model inputs. All values are engineering estimates as described in Section 5.

Table 11. *Consolidated one-third-octave transmission-loss schedules for all three cases (dB).*

Freq, Hz	Case 1	Case 2	Case 3
50	0.00	30.20	17.00
63	0.00	24.80	18.00
80	0.00	25.10	20.00
100	1.63	16.50	22.00
125	3.57	23.50	24.00
160	5.71	24.00	27.00
200	7.65	25.60	29.00
250	9.59	24.60	32.00
315	11.59	26.90	34.00
400	13.67	28.00	37.00
500	15.61	29.30	39.00
630	17.61	30.60	41.00
800	19.69	31.80	43.00
1000	21.63	32.30	45.00
1250	23.57	30.10	46.00
1600	25.71	25.50	47.00
2000	27.65	27.10	49.00
2500	29.59	34.00	50.00
3150	31.59	37.70	51.00
4000	33.67	39.80	53.00
5000	35.61	43.70	54.00
6300	37.61	46.50	55.00
8000	39.69	49.00	56.00
10000	41.63	51.00	57.00

Appendix B Full Octave-Band Sound-Pressure Levels

Tables 12 to 14 give the octave-band sound-pressure levels at the source-room receiver R1 and the receiving-room receiver R4 for each case, together with the band reduction ($R1 - R4$).

Table 12. Case 1 — octave-band sound-pressure levels at R1 and R4 with band reduction.

Octave band, Hz	R1 (source), dB	R4 (receiving), dB	Reduction R1 – R4, dB
63	54.8	51.9	2.9
125	54.8	49.5	5.3
250	56.4	44.4	12.0
500	60.1	39.8	20.3
1000	51.9	26.3	25.6
2000	45.6	13.4	32.2
4000	41.3	2.1	39.2
8000	41.0	–5.0	46.0

Table 13. Case 2 — octave-band sound-pressure levels at R1 and R4 with band reduction.

Octave band, Hz	R1 (source), dB	R4 (receiving), dB	Reduction R1 – R4, dB
63	54.2	25.6	28.6
125	54.7	32.9	21.8
250	56.4	28.1	28.3
500	60.1	25.9	34.2
1000	51.9	16.4	35.5
2000	45.6	13.1	32.5
4000	41.3	–4.3	45.6
8000	41.0	–14.1	55.1

Table 14. Case 3 — octave-band sound-pressure levels at R1 and R4 with band reduction.

Octave band, Hz	R1 (source), dB	R4 (receiving), dB	Reduction R1 – R4, dB
63	54.2	33.6	20.6
125	54.7	28.9	25.8
250	56.4	22.5	33.9
500	60.1	16.4	43.7
1000	51.9	3.2	48.7
2000	45.6	–7.7	53.3
4000	41.3	–17.0	58.3
8000	41.0	–21.5	62.5

Appendix C Raw ODEON Model Screenshot

Figure 13 reproduces the original, un-annotated ODEON model view for reference. The annotated version used in the main report is Figure 2.

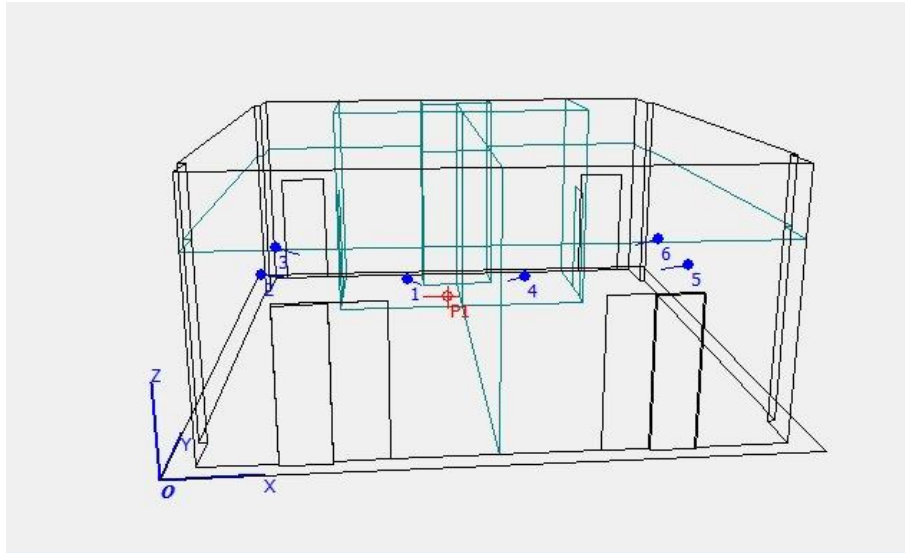


Figure 13. Raw ODEON model screenshot showing the two-room geometry, the source $P1$ and the receiver positions.

Appendix D Assumptions and Limitations

Table 15 summarises the key assumptions and their status, and the numbered list that follows states the limitations of the study in full.

Table 15. *Summary of key assumptions and their status.*

Item	Basis	Status
Case 1 TL schedule	Mass-law estimate, 1 mm aluminium sheet	Estimated input
Case 2 TL schedule	Aluminium–rockwool–aluminium sandwich	Preliminary input
Case 2 core thickness	100 mm rockwool core	Assumed value
Case 3 TL schedule	Double-leaf insulated stud partition	Conservative estimate
Source-room level	58.9 dB(A) at R1, identical for all cases	Model condition
Receiver positions	R1 source room; R4 receiving room	Fixed for all cases
Flanking transmission	Adjoining-structure paths	Not represented

1. The transmission-loss schedules used for all three cases are model inputs and engineering estimates, not laboratory-certified data.
2. The Case 1 schedule is a mass-law estimate for a thin aluminium sheet.
3. The Case 2 schedule is preliminary and assumes a 100 mm rockwool core; the core thickness is an assumed value.
4. The Case 3 schedule is a conservative engineering estimate for a double-leaf insulated stud partition.
5. The study is a comparative simulation and does not constitute laboratory or field certification of any assembly; no claim of certified sound insulation is made.
6. Before construction, the selected assembly should be confirmed against manufacturer test data and the relevant sound-insulation standards.
7. Flanking transmission along adjoining structures and the effects of workmanship and junction detailing are not represented in the model.
8. The results describe the modelled configuration and the specific receiver positions analysed; other receiver positions or room layouts may give different values.